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Strong Duality in Nonconvex Quadratic Optimization with Two Quadratic Constraints

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Abstract

We consider the problem of minimizing an indefinite quadratic function subject to two quadratic inequality constraints. When the problem is defined over the complex plane we show that strong duality holds and obtain necessary and sufficient optimality conditions. Using the results in the complex case and a result on the connection between the image of the real and complex spaces under a quadratic mapping, we are able to derive a condition that ensures strong duality even in the case of real variables. Preliminary numerical simulations show that for random instances of the extended trust region subproblem, the sufficient condition hardly never fails.

1 Introduction

In this paper we consider quadratic minimization problems with two quadratic constraints both in the real and in the complex domain:

$$(Q2P_{\mathbb{C}}) \qquad \min_{\boldsymbol{z}\in\mathbb{C}^n}\{f_3(\boldsymbol{z}) : f_1(\boldsymbol{z}) \geq 0, f_2(\boldsymbol{z}) \geq 0\}, \tag{1}$$

$$(Q2P_{\mathbb{R}}) \qquad \min_{\boldsymbol{x}\in\mathbb{R}^n}\{f_3(\boldsymbol{x}) : f_1(\boldsymbol{x}) \geq 0, f_2(\boldsymbol{x}) \geq 0\}. \tag{2}$$

In the real case each function $f_j : \mathbb{R}^n \to \mathbb{R}$ is defined by $f_j(\boldsymbol{x}) = \boldsymbol{x}^T\boldsymbol{A}_j\boldsymbol{x} + 2\boldsymbol{b}_j^T\boldsymbol{x} + c_j$ with $\boldsymbol{A}_j = \boldsymbol{A}_j^T \in \mathbb{R}^{n\times n}, \boldsymbol{b}_j \in \mathbb{R}^n$ and $c_j \in \mathbb{R}$. In the complex setting, $f_j : \mathbb{C}^n \to \mathbb{R}$ is given by $f_j(\boldsymbol{z}) = \boldsymbol{z}^*\boldsymbol{A}_j\boldsymbol{z} + 2\Re(\boldsymbol{b}_j^*\boldsymbol{z}) + c_j$, where $\boldsymbol{A}_j$ are Hermitian matrices, *i.e.*, $\boldsymbol{A}_j = \boldsymbol{A}_j^*$, $\boldsymbol{b}_j \in \mathbb{C}^n$ and $c_j \in \mathbb{R}$. The condition $\boldsymbol{A}_j = \boldsymbol{A}_j^*$ can be written as $\Re(\boldsymbol{A}_j)^T = \Re(\boldsymbol{A}_j)$ and $\Im(\boldsymbol{A}_j)^T = -\Im(\boldsymbol{A}_j)$.

The problem $(Q2P_{\mathbb{R}})$ appears as a subproblem in some trust region algorithms for constrained

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